```
> restart;#The monopole RadioFrequency Planck's "constant"
> Digits:=25;
```

$$Digits := 25 \tag{1}$$

```
> E(nu):=Em+E0*ln(1+2*Tq*A*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(2\, Tq\, A\, \nu + 1) + C\, \nu^2 \tag{2}$$

```
> Tq:=1.765*10^(-19);mu:=1.25663706212*10^(-6);B:=10^13;k:=6.75*10^4;rP:=
  1.5346982640656*10^(-18);ec:= 1.602176634*10^(-19);
```

$$Tq := 1.765000000000000000000000 \times 10^{-19}$$

$$\mu := 1.256637062120000000000000 \times 10^{-6}$$

$$B := 10000000000000$$

$$k := 67500.00$$

$$rP := 1.534698264065600000000000 \times 10^{-18}$$

$$ec := 1.602176634000000000000000 \times 10^{-19} \tag{3}$$

```
> A0:=64/3*Tq;
```

$$A0 := 3.765333333333333333333333 \times 10^{-18} \tag{4}$$

```
> E(nu):=Em+E0*ln(1+A*3.7653333333333333333*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 3.765333333333333333300000 \times 10^{-18} A\, \nu) + C\, \nu^2 \tag{5}$$

```
> E1:=subs(nu=1150.834772*10^6,E(nu));
```

$$E1 := Em + E0 \ln(1 + 4.333276528170666666628306 \times 10^{-9} A) + 1.324420672444291984000000 \times 10^{18} C \tag{6}$$

```
> E2:=subs(nu=575.417386*10^6,E(nu));
```

$$E2 := Em + E0 \ln(1 + 2.166638264085333333314153 \times 10^{-9} A) + 3.311051681110729960000000 \times 10^{17} C \tag{7}$$

```
> E3:=subs(nu=30.411865*10^9,E(nu));
```

$$E3 := Em + E0 \ln(1 + 1.145108090133333333323196 \times 10^{-7} A) + 9.248815327782250000000000 \times 10^{20} C \tag{8}$$

```
> solve({E1=3.259,E2=1.6295});
```

$$\{A = A,\ C = -1.006729478838919463670757 \times 10^{-18} E0 \ln(1. + 4.333276528170666666628306 \times 10^{-9} A) \tag{9}$$
$$+ 1.006729478838919463670757 \times 10^{-18} E0 \ln(1. + 2.166638264085333333314153 \times 10^{-9} A)$$
$$+ 1.640465685768019266051499 \times 10^{-18},\ E0 = E0,\ Em = -1.333333333333333333333333\, E0 \ln(1.$$
$$+ 2.166638264085333333314153 \times 10^{-9} A) + 0.333333333333333333333333\, E0 \ln(1.$$
$$+ 4.333276528170666666628306 \times 10^{-9} A) + 1.086333333333333333333333\}$$

```
> sol:=solve({E1=3.259,E2=1.6295,E3=1.6295},{Em,E0,C,A});
```

$$sol := \{A = A,\ C = (0.13579166666666666666666667\ (-1. \ln(1. + 1.145108090133333333323196 \tag{10}$$
$$\times 10^{-7} A) + \ln(1. + 2.166638264085333333314153 \times 10^{-9} A)))/(-8.2776292027768249$$
$$\times 10^{16} \ln(1. + 1.145108090133333333323196 \times 10^{-7} A) + 7.7045868967509493917 \times 10^{19} \ln(1.$$
$$+ 4.333276528170666666628306 \times 10^{-9} A) - 7.6963092675481725668 \times 10^{19} \ln(1.$$

$$+ 2.16663826408533333331415 3 \times 10^{-9} A\Big)\Big), E0 = \Big(1.25546243482556720337751 5 \times 10^{20}\Big) \Big/ \Big($$

$$-8.2776292027768249 \times 10^{16} \ln\Big(1. + 1.14510809013333333332319 6 \times 10^{-7} A\Big)$$

$$+ 7.7045868967509493917 \times 10^{19} \ln\Big(1. + 4.33327652817066666662830 6 \times 10^{-9} A\Big)$$

$$- 7.6963092675481725668 \times 10^{19} \ln\Big(1. + 2.16663826408533333331415 3 \times 10^{-9} A\Big)\Big), Em$$

$$= \Big(0.54316666666666666666666667 \Big(-4.62109661221001427004 \times 10^{20} \ln\Big(1.$$
$$+ 2.16663826408533333331415 3 \times 10^{-9} A\Big) - 1.65552584055536498 \times 10^{17} \ln\Big(1.$$
$$+ 1.14510809013333333332319 6 \times 10^{-7} A\Big) + 2.31137606902528481751 \times 10^{20} \ln\Big(1.$$
$$+ 4.33327652817066666662830 6 \times 10^{-9} A\Big)\Big)\Big) \Big/ \Big(-8.2776292027768249 \times 10^{16} \ln\Big(1.$$
$$+ 1.14510809013333333332319 6 \times 10^{-7} A\Big) + 7.7045868967509493917 \times 10^{19} \ln\Big(1.$$
$$+ 4.33327652817066666662830 6 \times 10^{-9} A\Big) - 7.6963092675481725668 \times 10^{19} \ln\Big(1.$$
$$+ 2.16663826408533333331415 3 \times 10^{-9} A\Big)\Big)\Big\}$$

```
> A:=4197470;evalf(eval(Em,sol));#Em
```

$$A := 4197470$$

$$-0.06774149155738240034443890 \quad (11)$$

```
> Em:=-0.06774149155738240034443900;
```

$$Em := -0.06774149155738240034443900 \quad (12)$$

```
> evalf(eval(E0,sol));#E0=190.3595555260865435980605;
```

$$190.35955552608654359806 36 \quad (13)$$

```
> E0:=190.3595555260865435980605;
```

$$E0 := 190.35955552608654359806 05 \quad (14)$$

```
> E0m:=evalf(k*Pi*rP^3*B^2/(2*mu)/ec);evalf((E0m-E0)/E0);#E0 from k*Pi*rP^3*
  B^2/(2*mu) with B=1e13 T, versus E0 of the fit
```

$$E0m := 190.3585541501417056546608$$

$$-5.260444856947129063841797 \times 10^{-6} \quad (15)$$

```
> evalf(eval(C,sol));#C
```

$$-7.894636227216279440136258 \times 10^{-20} \quad (16)$$

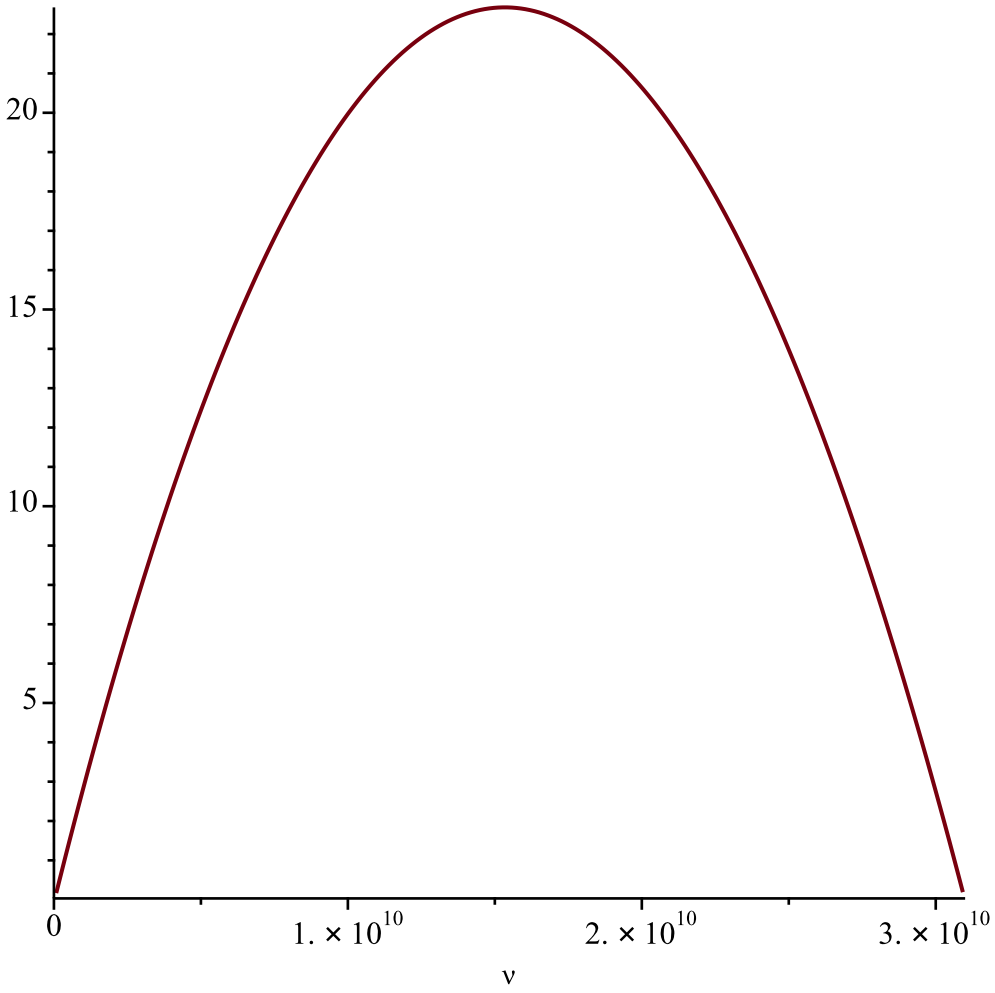
```
> C:=-7.894636227216279440136117*10^(-20);
```

$$C := -7.894636227216279440136117 \times 10^{-20} \quad (17)$$

```
> E(nu):=Em+E0*ln(1+A*3.765333333333333333*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := -0.06774149155738240034443900 + 190.35955552608654359806 05 \ln\Big(1 \quad (18)$$
$$+ 1.58048737066666666666526 75 \times 10^{-11} \nu\Big) - 7.894636227216279440136117 \times 10^{-20} \nu^2$$

```
> plot(E(nu),nu=10^6..3.1*10^10);
```




```
> evalf(subs(nu=2*5.754173859*10^8,E(nu)));evalf(subs(nu=2*6.338671958*10^8,
  E(nu)));evalf(subs(nu=2*6.974105195*10^8,E(nu)));
```

$$3.258999999445369023285367$$

$$3.581788888344925176349726$$

$$3.929550309031864693074719 \quad \textbf{(19)}$$

```
> evalf(subs(nu=0,E(nu)));evalf(subs(nu=22.54*10^6,E(nu)));evalf(subs(nu=
  15.7*10^9,E(nu)));#22.67eV @ 15.7Ghz in the range [22.5Mhz-30.99Ghz]
```

$$-0.06774149155738240034443900$$

$$0.00002036424827068047520386$$

$$22.666279769661539670743 43 \quad \textbf{(20)}$$

```
> #Gauss's law: point charge q at the centre of a cylinder of radius Rc and
  height H=Rc (fluxes in units of q/epsilon)
> Phicaps:=2*int(d*r/(2*(r^2+d^2)^(3/2)),r=0..Rc) assuming d>0,Rc>0;Philat:=
  int(Rc^2/(2*(Rc^2+z^2)^(3/2)),z=-d..d) assuming d>0,Rc>0;
```

$$Phicaps := \frac{\sqrt{Rc^2 + d^2} - d}{\sqrt{Rc^2 + d^2}}$$

$$Philat := \frac{d}{\sqrt{Rc^2 + d^2}} \quad \textbf{(21)}$$

```
> caps:=simplify(subs(d=Rc/2,Phicaps)) assuming Rc>0;lat:=simplify(subs(d=
  Rc/2,Philat)) assuming Rc>0;simplify(caps+lat);evalf(caps);evalf(lat);
```

$$caps := \frac{\left(\sqrt{5}-1\right)\sqrt{5}}{5}$$

$$lat := \frac{\sqrt{5}}{5}$$

$$1$$

$$0.55278640450004206071816 54$$

$$0.44721359549995793928183 48 \quad \textbf{(22)}$$

```
> E0mw:=424.9842199315626666597335;evalf(E0mw/E0);evalf(1/lat);#sqrt(5)=
  Phi_sphere/Phi_lateral
```

$$E0mw := 424.9842199315626666597335$$

$$2.232534210100756227358277$$

$$2.236067977499789696409174 \quad \textbf{(23)}$$

```
> evalf(sqrt(E0mw/E0));evalf(5^(1/4));evalf(sqrt(E0mw/E0)/5^(1/4)-1);
  #B_mw/B_m and 5^(1/4)
```

$$1.494166727678258878769103$$

$$1.495348781221220541911899$$

$$-0.000790486846818642783 3579 \quad \textbf{(24)}$$

```
> solve(-0.06774149155738240034443900 + 190.35955552608654359806 05*ln(1 +
  1.58048737066666666666526 75*10^(-11)*nu) - 7.89463622721627944013 6117*10^
  (-20)*nu^2 -3.95=0,nu);
```

Warning, solutions may have been lost

```
> evalf(subs(nu=2*7.01*10^8,E(nu)));E(2*7.01*10^8);1239.8/336.8;
```

$$3.949096842314479833687537$$

$$E\left(1.40200000000 \times 10^{9}\right)$$

$$3.681116389548693586698337 \quad \textbf{(25)}$$

```
> evalf(subs(nu=2*6.5196*10^8,E(nu)));E(2*6.5196*10^8);
```

$$3.681142334365752350938675$$

$$E\left(1.3039200000000 \times 10^{9}\right) \quad \textbf{(26)}$$